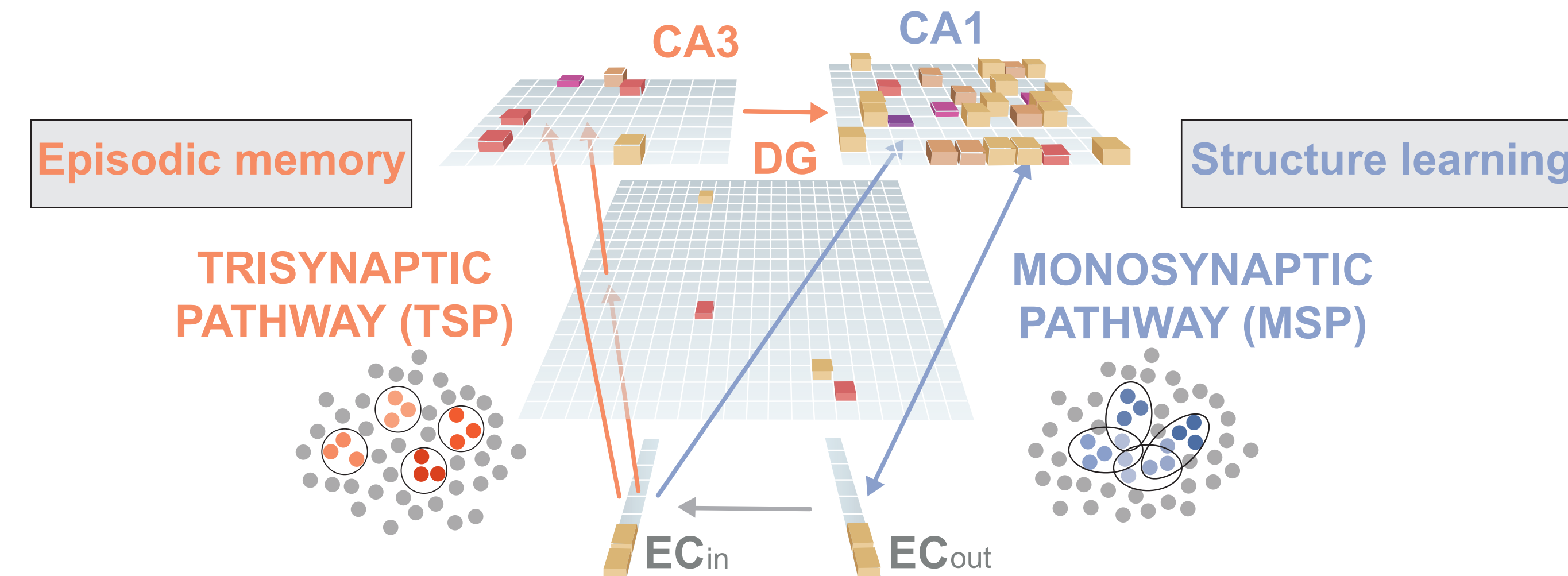


## Introduction

The hippocampus is critical for both **episodic memory** and **structure learning**, functions that benefit from opposing styles of representation. C-HORSE<sup>1</sup> proposes that the hippocampus (HPC) contains an anatomical dissociation to support these functions:



**How does the brain deploy the most useful representation for the current task?**

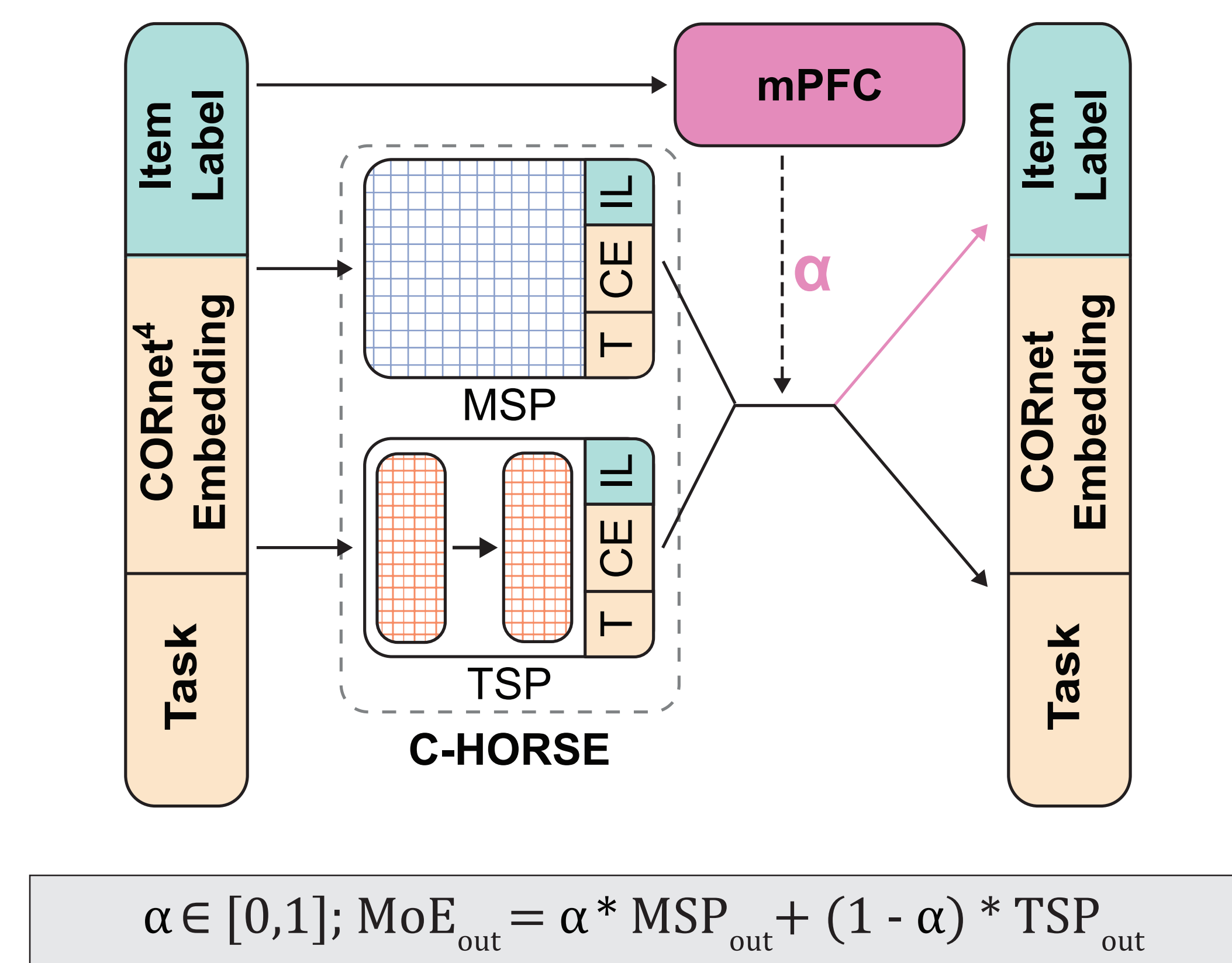
Medial prefrontal cortex (mPFC) has been implicated in exerting control over hippocampal function<sup>2</sup> and is involved in both episodic and structure learning tasks. It could play a role in deploying task-adaptive hippocampal representations.

We provide an account of these interactions using a stimulus-computable **mixture-of-experts system (MoE)**<sup>3</sup>, where an **mPFC-inspired output gating network** learns to control the **task-specific deployment of hippocampal representations**.

## Model Architecture

Our model involves a C-HORSE module, including **MSP** and **TSP** expert networks. An **mPFC** gating network produces mixture weights to balance their contributions to the final MoE system output.

The system is set up as an autoencoder: reconstructing input task signal, item embedding, and item label on the output heads.

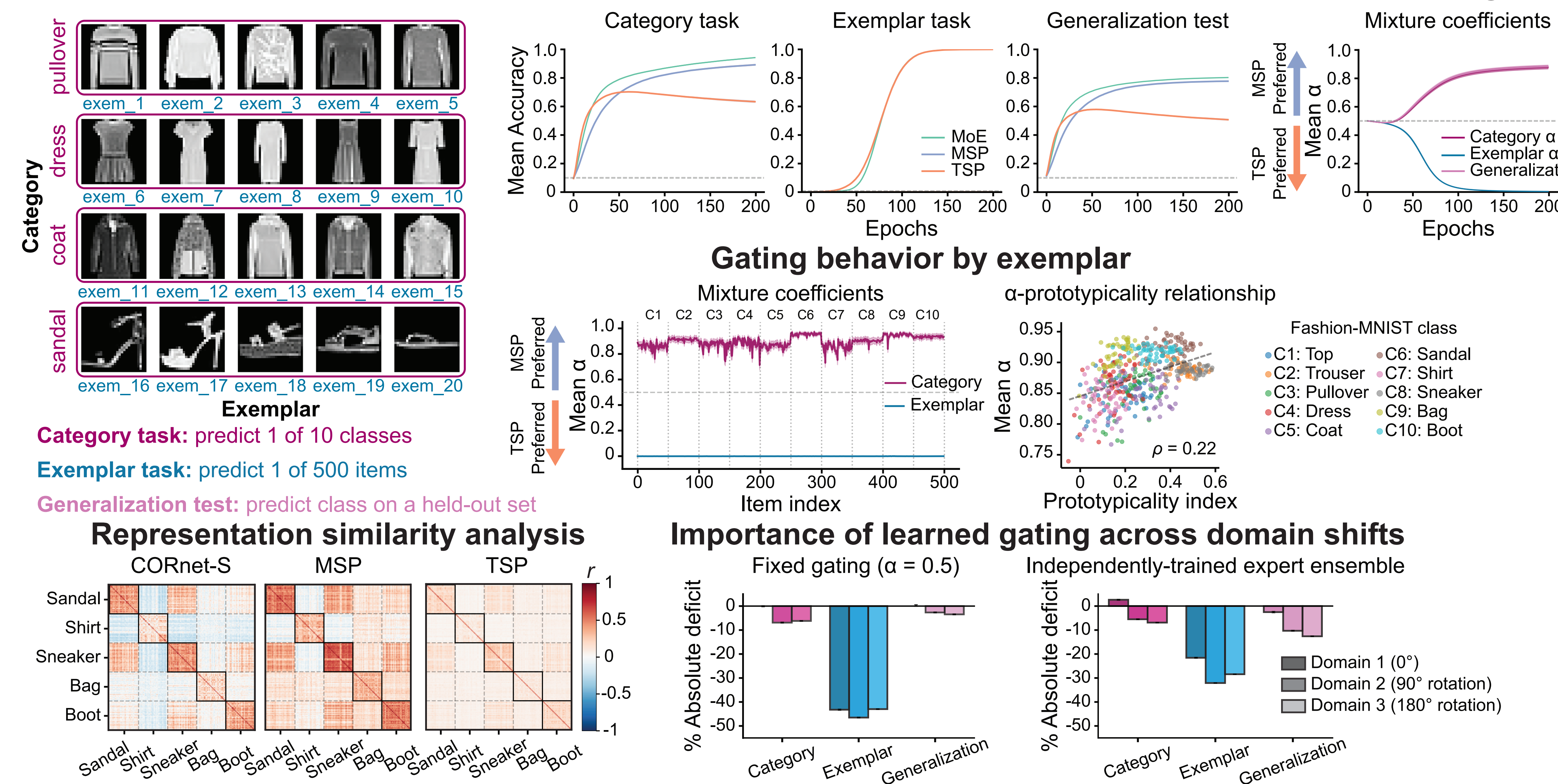


$$\alpha \in [0,1]; \text{MoE}_{\text{out}} = \alpha * \text{MSP}_{\text{out}} + (1 - \alpha) * \text{TSP}_{\text{out}}$$

|                     | MSP   | TSP   | mPFC  |
|---------------------|-------|-------|-------|
| Parameter count     | Small | Large | Small |
| Learning rate       | Low   | High  | Low   |
| Sparsity/Inhibition | None  | High  | None  |

## Simulation 1: Category learning proof-of-concept

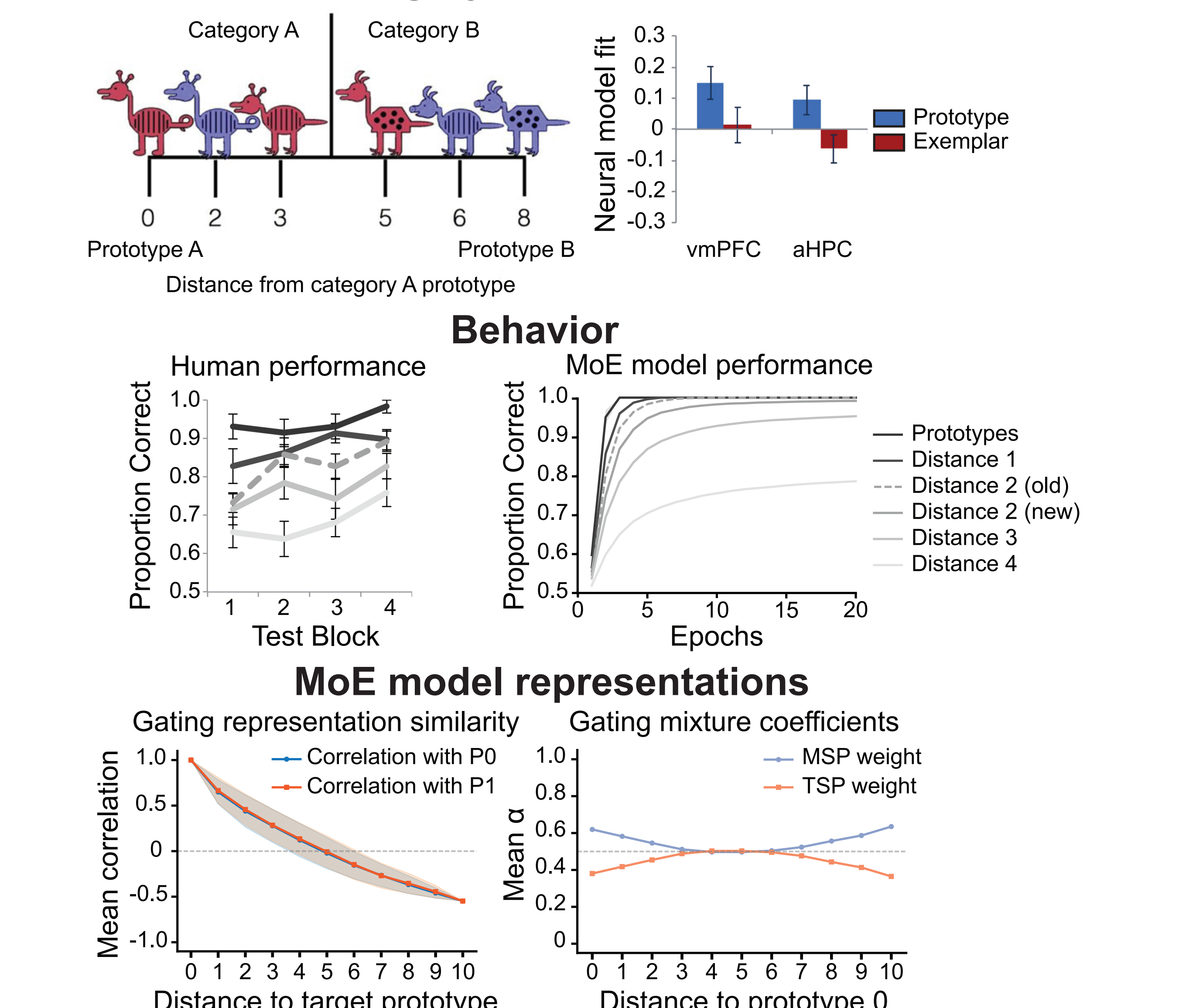
Fashion-MNIST<sup>5</sup> subset (N=500)



Experts exhibit a double dissociation in specialization, which the mPFC gate learns to leverage. Preferential TSP engagement for exceptional items is consistent with prior modeling and empirical findings<sup>6,7,8</sup>.

## Simulation 3: Structured representations within mPFC

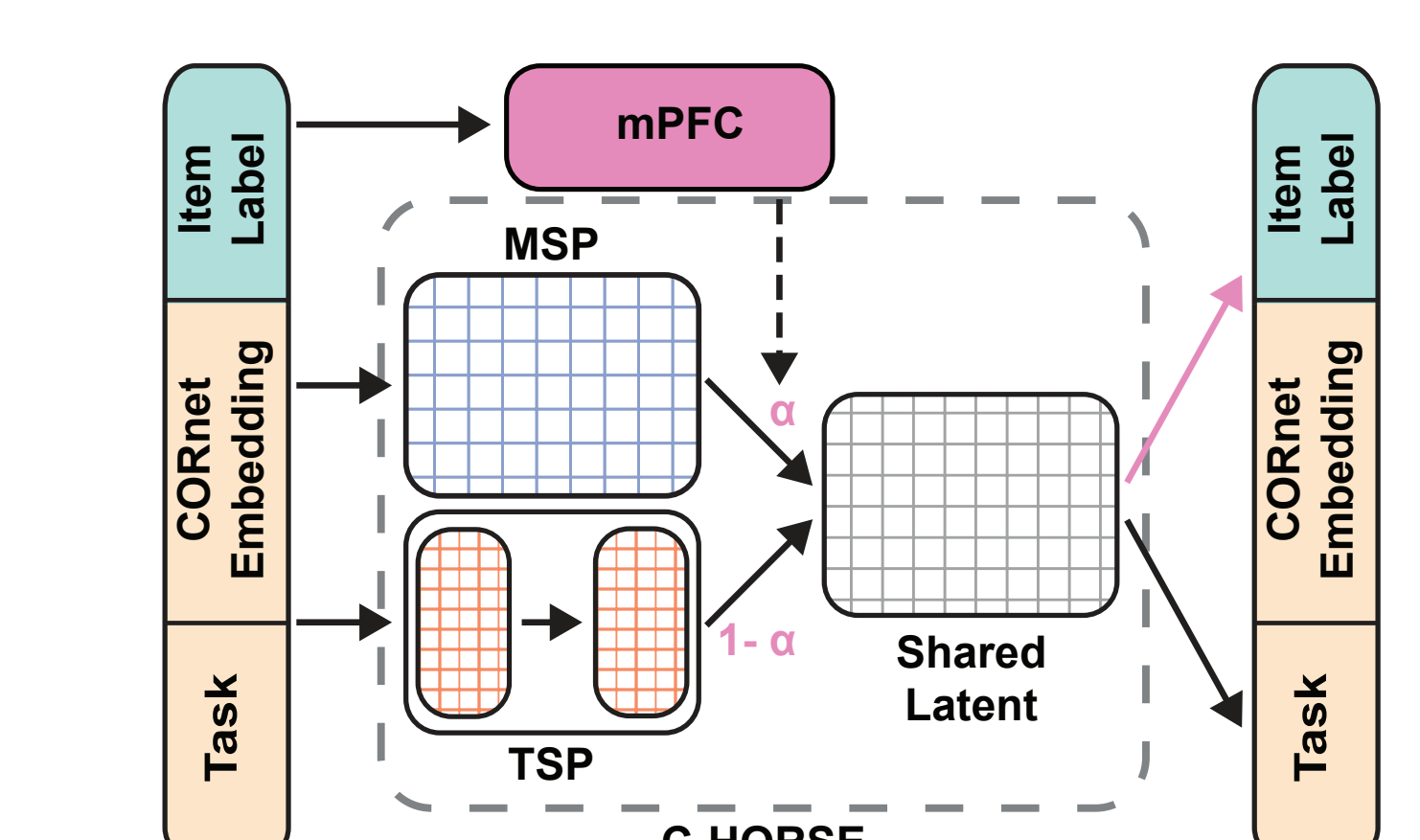
Bowman et al. category structure & fMRI model fits<sup>11</sup>



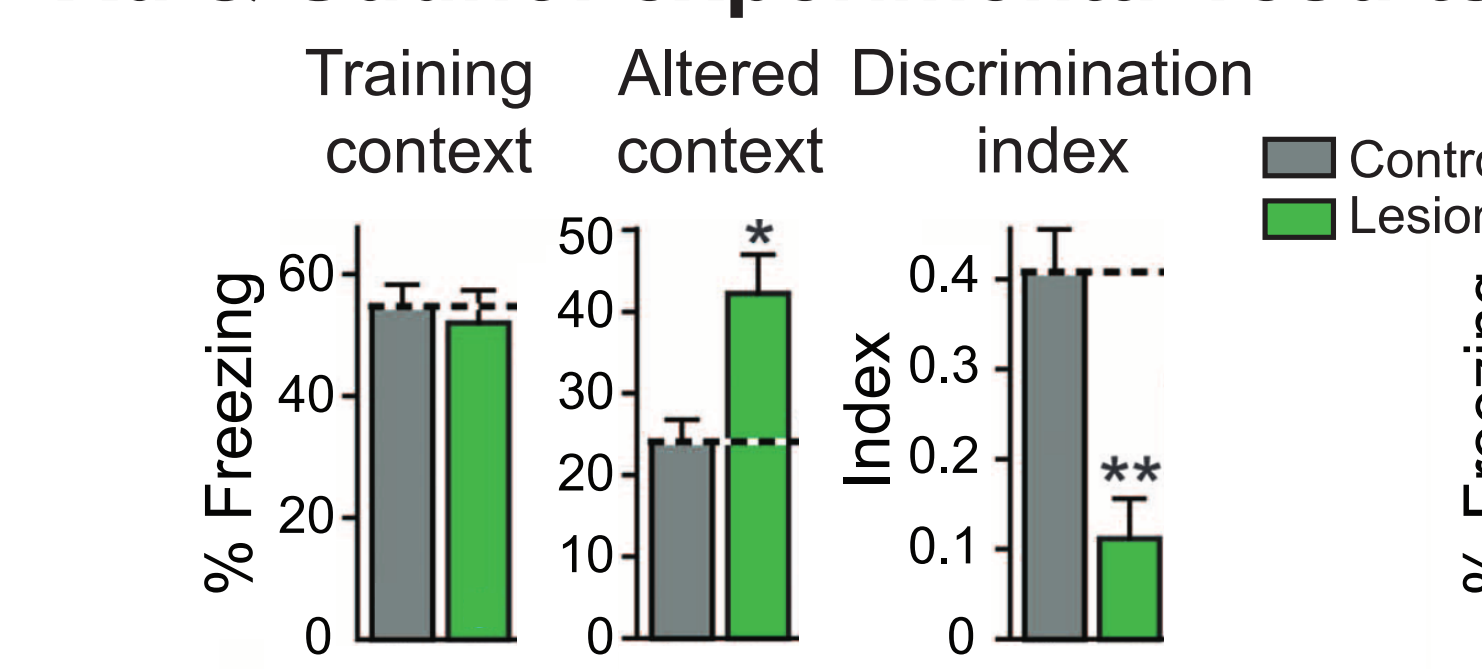
Gating network develops graded prototype sensitivity, demonstrating that structured representations in mPFC can emerge in service of control.

## Simulation 2: mPFC control over memory specificity

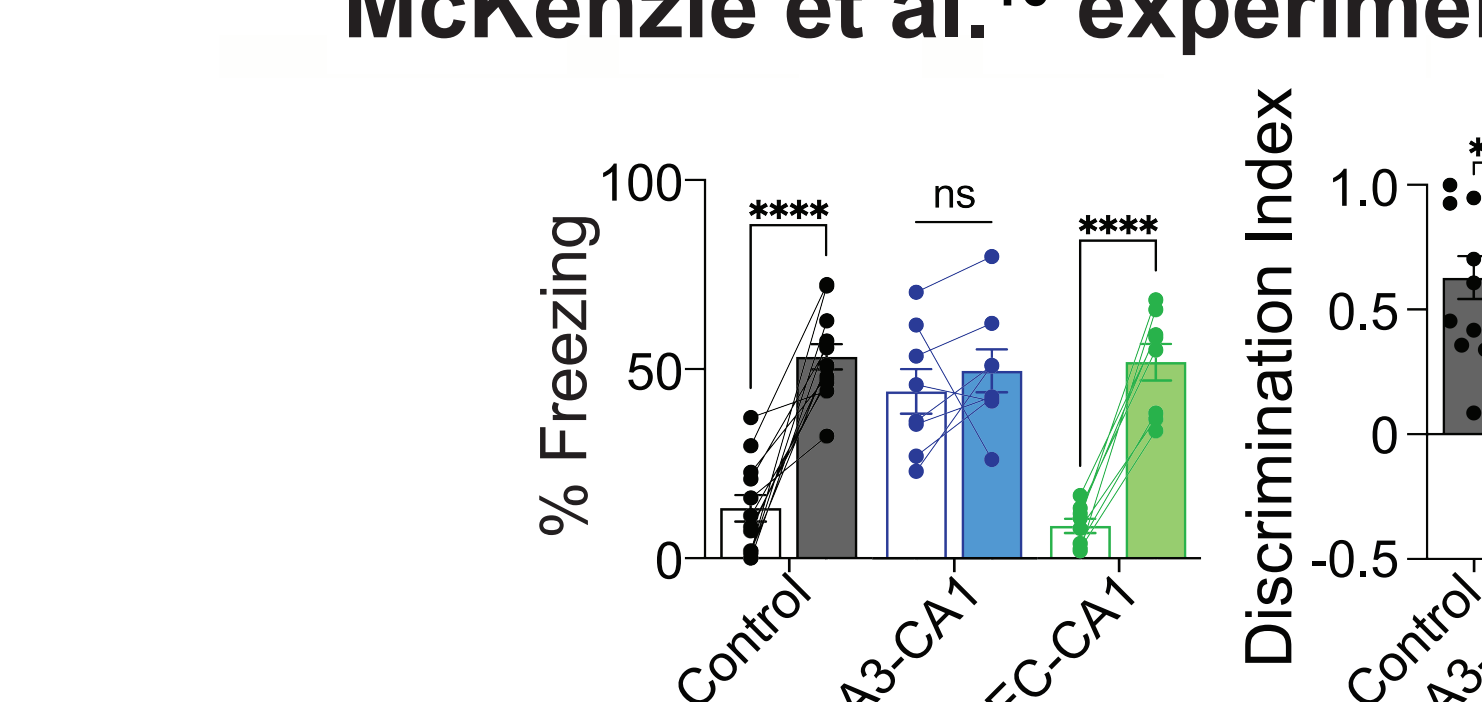
Shared latent model architecture



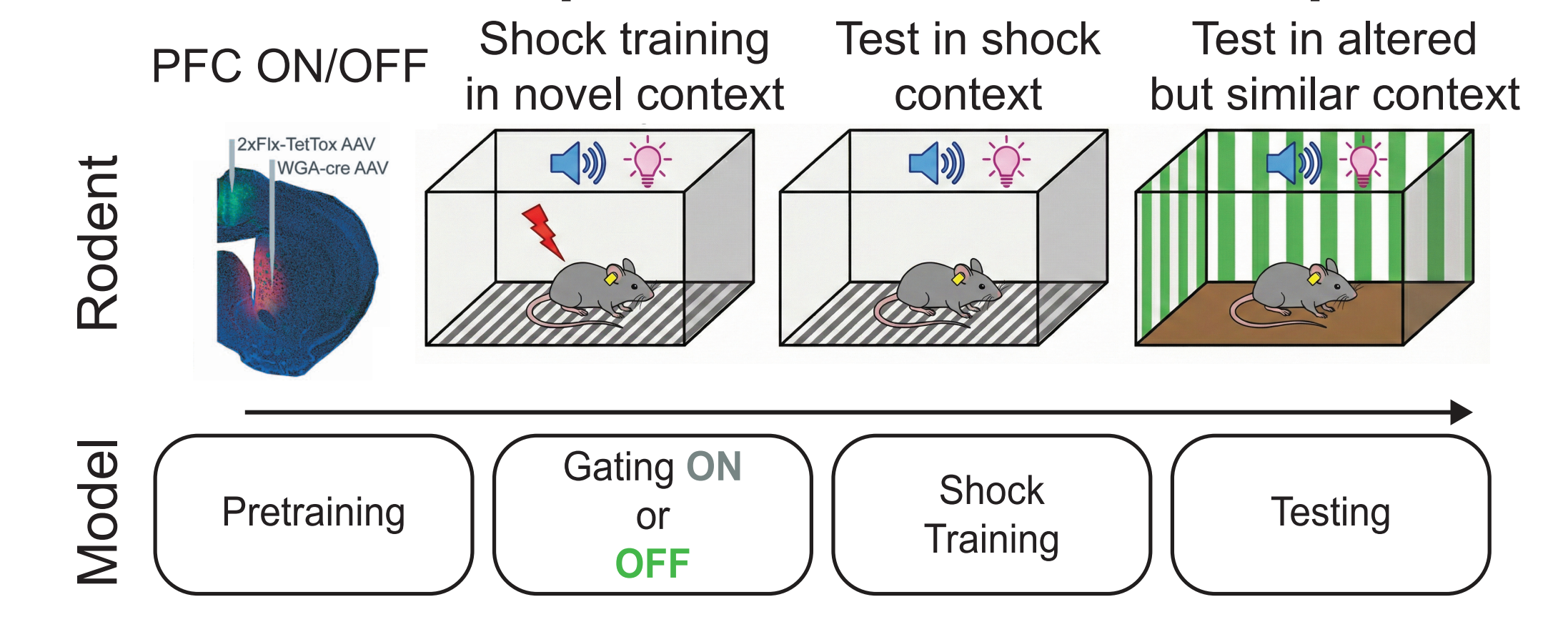
Xu & Südhof experimental results



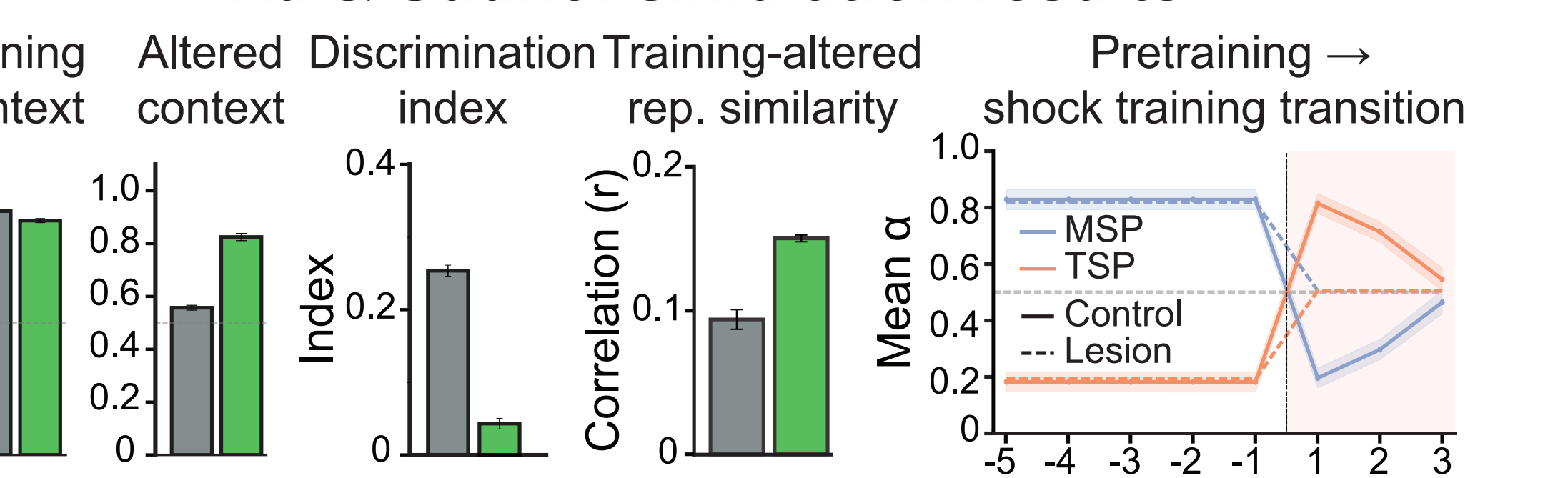
McKenzie et al.<sup>10</sup> experimental results



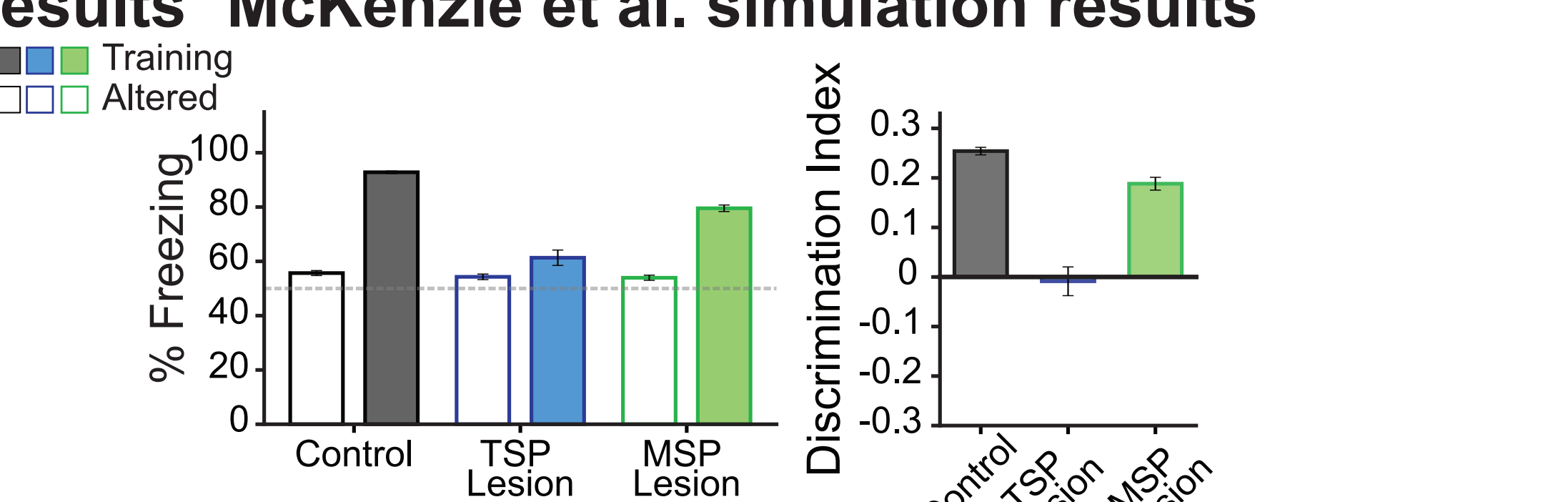
Xu & Südhof<sup>9</sup> experimental & simulation protocol



Xu & Südhof simulation results



McKenzie et al. simulation results



mPFC control engages TSP for the novel shock context, enabling behavioral and representational specificity. Loss of gating attenuates dynamic TSP engagement, and accounts for lesion-driven overgeneralization.

## Take Home

- Adaptive mPFC gating over complementary hippocampal representations provides a unifying mechanistic account of how the brain arbitrates between episodic and structure memory.
- Impaired gating disrupts overall performance and produces a loss of representational specificity, accounting for mPFC lesion-driven overgeneralization observed in rodent contextual fear conditioning.
- The gating network's representational geometry tracks category prototypicality, raising the possibility that abstraction in mPFC may emerge in service of control.
- Testable predictions: (i)  $\alpha$  should map onto an axis in mPFC activity space—tracking prototypicality and predicting MSP engagement; (ii) mPFC damage should produce larger deficits after domain or rule changes, and on tasks requiring fine-grained episodic discrimination.

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